

Comparing Variable Selection Procedures Using Simulation

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Introduction

Variable selection is a tool widely used when modelling to find parsimonious and predictive models. Many methods are available for variable selection with different use cases, strengths, and weaknesses. Because variable selection may sometimes be of use, for example for predictive modelling, it is of interest to investigate how these methods compare with varying correlation structures and sample sizes. For this project, we will compare backwards selection using p-values, AIC, and BIC, as well as regression using LASSO and elastic net, using simulated datasets with known correlations between the variables, and performance metrics including true and false positive rates, type I and II error rates of variables selected (using $\alpha=.05$), confidence interval coverage, and bias. We want to test whether these methods perform similarly or whether the error rate, bias, and coverage differ across the models, with 2 sample size cases, and 3 correlation structure cases.

Methods

We are comparing 7 methods for variable selection. Backwards selection using p-values with $\alpha = .1$ removes variables one model fit at a time, removing the term with the highest p-value until all included terms have $p < \alpha$, a threshold set by the user. Here we used .1 as a stricter penalty than AIC (which corresponds to using $\alpha = .157$). Backwards selection using AIC or BIC perform similarly, stopping once a model's AIC or BIC is not improved (decreased) by the exclusion of an additional variable. We also implemented LASSO (Least Absolute Shrinkage and Selection Operator) regression which shrinks unnecessary or non-predictive coefficients towards zero. We considered two penalty term options, either the λ_{min} such that λ_{min} minimizes the mean cross-validated error, and the λ_1 value 1 standard deviation larger than λ_{min} . These two choices are conventionally used, since the 1 standard error case provides a parsimonious model with error no more than 1 standard error that of λ_{min} model, since minimizing the error may lead to overfitting (Krstajic 2014). We used these same λ options for elastic-net, which we implemented using $\alpha = .5$, because we wanted to implement elastic net as distinctively from ridge regression ($\alpha = 0$) and LASSO ($\alpha = 1$) as possible. Elastic net balances the penalization methods of LASSO and Ridge Regression, which is used to handle multicollinearity between variables. The p-value method was implemented using the `ols_step_backward_p()` function in the `olsrr` package in R, AIC and BIC using the `step()` function with $k=2$ and $k = \log(n)$ respectively, and the LASSO and E-net methods were implemented using the above described parameters. All models were fit with an intercept term, but it was not considered in results comparisons.

Table 1: Summary of methods compared

Method	Functions	Package	Parameters
P-values	<code>ols_step_backward_p()</code>	<code>olsrr</code>	$p=.1$
AIC	<code>step()</code>	<code>stats (base)</code>	$k=2$
BIC	<code>step()</code>	<code>stats (base)</code>	$k = \log(n)$
LASSO (1)	<code>glmnet, cv.glmnet</code>	<code>glmnet</code>	$\lambda = \lambda_{min}, \alpha = 1$
LASSO (2)	<code>glmnet, cv.glmnet</code>	<code>glmnet</code>	$\lambda = \lambda_{1se}, \alpha = 1$
E-net (1)	<code>glmnet, cv.glmnet</code>	<code>glmnet</code>	$\lambda = \lambda_{min}, \alpha = .5$
Enet (2)	<code>glmnet, cv.glmnet</code>	<code>glmnet</code>	$\lambda = \lambda_{1se}, \alpha = .5$

Because we were interested in both coverage and bias in our simulations, we used two approaches from Morris 2019 to calculate needed n_{sim} based on these two measures. Attempting these two methods provided divergent results, so we chose to use $n_{sim} = 10000$ to ensure adequate power, as this benchmark is often used and recommended (Morris 2019). For each iteration, we generated 6 datasets using the `hdrm` package in R, with sample sizes 250 and 500, as well as exchangeable correlation structure within the design matrix with $\rho = 0$ (no correlation), $\rho = .35$, and $\rho = .7$. All variables were generated as Gaussian, with the predictors following a multivariate normal distribution with mean 0, variance 1, and covariance determined by correlation case. For each dataset, 5 of the 20 variables in the design matrix had a true association with the outcome vector y , with true coefficients for variables V_1, V_2, V_3, V_4, V_5 as $1/6, 1/3, 1/2, 2/3$, and $5/6$ respectively. With the exchangeable correlation structure, V_1 through V_{20} are assumed to have a between-variable correlation of ρ , and sampled accordingly. We then performed variable selection for each of the 6 cases ($N=250, \rho = 0, .35, .7$, and $N=500, \rho = 0, .35, .7$) using the 7 described methods. Multiple linear regression models using least squares were refit in R for y regressed on the selected variables from each method. From these refit models, we retained information on the bias and coverage of the coefficients vs. their true values, the amount of times that the models included V_1 through V_5 , the variables with true associations, the amount of times the models included V_6 through V_{20} (the null variable group), and how often type 1 and type 2 errors were committed. Bias was measured as the average difference between the coefficient estimate and the true coefficient specified above, with the true coefficient for V_6 through V_{20} equal to 0. For the true effect variables, coverage was defined as the proportion of iterations where the model's Wald CI for the estimate of interest included the true effect. If a true effect

variable was not included in the model, the variable was treated as having estimate 0 with a confidence interval only containing 0. The true positive rate (TPR) and false positive rate (FPR) were reported at the model level; a model with V_1 through V_5 included in the model was a true positive case, and a model with at least one of the variables V_6 through V_{20} was a false positive. We calculated type 1 error rates two ways. The first way was at the model level, e.g., how often was at least one variable in the null group included in the regression model with a significant p-value (using $\alpha = .05$ as a threshold since this is often used in medical studies) and also as the average number of type 1 errors committed by a modelling approach for a given case. Similarly, we investigated type 2 error at the model level and reported the average number of type 2 errors committed by modelling approach and case. Reporting the number of errors per model allows us to compare approaches at a more granular level since, for example, there is a difference between a model with 3 insignificant variables included and a model with 1, though both would be reported analogously at the model level. We also investigated type 2 error rates for the individual variables since we had a range of coefficient strengths across V_1 through V_5 . Reporting these measures this way allows us to understand model patterns more robustly since we can look at trends at both the model level and variable level. Both type 1 and type 2 error rates are handled as though no model selection had been done, which will inflate type 1 error rates and underestimate standard error (Shen 2024). Since this naive approach is often used in practice, it is beneficial to see how methods compare in this context. Simulation procedure was parallelized using doRNG package in R, and we used seed 23192319 to ensure reproducible results. Code for simulating the data and producing tables and plots is available at <https://github.com/jkebe2002/BIOS6624/tree/9c435c37471509d45ecdb37552575dbf7bd10cd7/P4>.

Results

Table 2: Bias and Coverage by Method, Variable, and Correlation Structure

		V01		V02		V03		V04		V05	
N	Method	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.	Bias	Cov.
rho = 0											
N = 250											
N=250	P-Value	-0.012	0.800	-0.001	0.944	-0.001	0.949	0.000	0.941	0.001	0.943
N=250	AIC	-0.007	0.854	-0.001	0.943	-0.001	0.947	0.000	0.939	0.001	0.942
N=250	BIC	-0.041	0.585	-0.001	0.946	-0.001	0.951	-0.001	0.944	0.001	0.944
N=250	LASSO1	-0.004	0.908	-0.003	0.943	-0.002	0.946	-0.002	0.939	-0.001	0.940
N=250	LASSO2	-0.047	0.562	-0.004	0.945	-0.003	0.951	-0.002	0.943	-0.001	0.944
N=250	ENET1	-0.028	0.694	-0.004	0.943	-0.003	0.949	-0.003	0.942	-0.002	0.945
N=250	ENET2	-0.002	0.929	-0.003	0.943	-0.002	0.947	-0.002	0.940	-0.001	0.941
N = 500											
N=500	P-Value	-0.001	0.948	-0.001	0.947	0.000	0.949	0.000	0.946	0.000	0.948
N=500	AIC	-0.001	0.948	-0.001	0.946	0.000	0.949	0.000	0.947	0.000	0.947
N=500	BIC	-0.010	0.865	-0.001	0.947	0.000	0.952	0.000	0.948	0.000	0.950
N=500	LASSO1	-0.001	0.948	-0.001	0.946	0.000	0.950	-0.001	0.946	0.000	0.946
N=500	LASSO2	-0.018	0.805	-0.001	0.948	0.000	0.951	0.000	0.948	0.000	0.949
N=500	ENET1	-0.008	0.890	-0.001	0.947	0.000	0.951	-0.001	0.946	0.000	0.950
N=500	ENET2	-0.001	0.948	-0.001	0.946	0.000	0.950	-0.001	0.947	0.000	0.947
rho = 0.35											
N = 250											
N=250	P-Value	-0.022	0.679	0.002	0.941	0.002	0.936	0.002	0.934	0.003	0.939
N=250	AIC	-0.013	0.751	0.002	0.941	0.001	0.935	0.001	0.935	0.002	0.939
N=250	BIC	-0.058	0.444	0.006	0.938	0.008	0.935	0.008	0.933	0.008	0.938
N=250	LASSO1	-0.023	0.923	-0.023	0.935	-0.024	0.932	-0.024	0.930	-0.024	0.932
N=250	LASSO2	-0.034	0.795	-0.024	0.934	-0.025	0.931	-0.025	0.927	-0.025	0.932
N=250	ENET1	-0.034	0.907	-0.034	0.931	-0.035	0.929	-0.035	0.927	-0.035	0.929
N=250	ENET2	-0.021	0.932	-0.022	0.937	-0.022	0.934	-0.023	0.933	-0.022	0.933
N = 500											
N=500	P-Value	-0.003	0.904	0.000	0.942	0.000	0.942	0.000	0.942	0.000	0.939
N=500	AIC	-0.001	0.928	0.000	0.942	0.000	0.943	0.000	0.943	0.000	0.940
N=500	BIC	-0.022	0.745	0.003	0.942	0.003	0.943	0.003	0.944	0.003	0.940
N=500	LASSO1	-0.016	0.934	-0.017	0.932	-0.017	0.935	-0.017	0.934	-0.017	0.933
N=500	LASSO2	-0.017	0.926	-0.016	0.935	-0.016	0.937	-0.016	0.935	-0.016	0.931
N=500	ENET1	-0.023	0.930	-0.025	0.928	-0.025	0.930	-0.024	0.928	-0.025	0.926
N=500	ENET2	-0.014	0.936	-0.016	0.936	-0.016	0.938	-0.015	0.937	-0.016	0.936
rho = 0.7											
N = 250											
N=250	P-Value	-0.047	0.415	-0.007	0.872	0.005	0.935	0.006	0.933	0.006	0.931
N=250	AIC	-0.031	0.512	-0.003	0.904	0.003	0.937	0.003	0.932	0.003	0.935
N=250	BIC	-0.092	0.201	-0.022	0.755	0.018	0.923	0.022	0.920	0.022	0.921
N=250	LASSO1	-0.040	0.811	-0.048	0.920	-0.049	0.924	-0.048	0.925	-0.048	0.923
N=250	LASSO2	-0.052	0.704	-0.058	0.911	-0.058	0.916	-0.057	0.918	-0.057	0.917
N=250	ENET1	-0.051	0.881	-0.059	0.920	-0.060	0.926	-0.059	0.927	-0.058	0.927
N=250	ENET2	-0.037	0.841	-0.045	0.922	-0.046	0.926	-0.045	0.927	-0.045	0.926
N = 500											
N=500	P-Value	-0.025	0.653	0.001	0.938	0.003	0.940	0.003	0.936	0.003	0.933
N=500	AIC	-0.016	0.728	0.001	0.937	0.001	0.941	0.002	0.936	0.002	0.934
N=500	BIC	-0.072	0.361	0.008	0.924	0.013	0.931	0.014	0.929	0.013	0.925
N=500	LASSO1	-0.031	0.918	-0.033	0.925	-0.033	0.930	-0.032	0.921	-0.032	0.921
N=500	LASSO2	-0.040	0.867	-0.040	0.919	-0.039	0.923	-0.038	0.914	-0.039	0.917
N=500	ENET1	-0.040	0.925	-0.041	0.925	-0.041	0.930	-0.040	0.923	-0.040	0.924
N=500	ENET2	-0.030	0.924	-0.031	0.927	-0.031	0.930	-0.030	0.924	-0.030	0.924

As shown in Table 2, a few trends persisted relating to bias and coverage. For example, coverage was worst for smaller sample sizes and for V_1 , the variable with weakest positive association with the outcome. However, in both these cases, LASSO using λ_{min} and elastic net using λ_1 both outperformed other methods. For variables V_2 through V_5 , coverage was generally at or around 93% but backwards-selection based methods underperformed in the case of lowest sample size and highest correlation. However, backwards-selection methods tended to have lower bias than the penalized regression methods. In fact, AIC tended to have the lowest bias compared to all other methods.

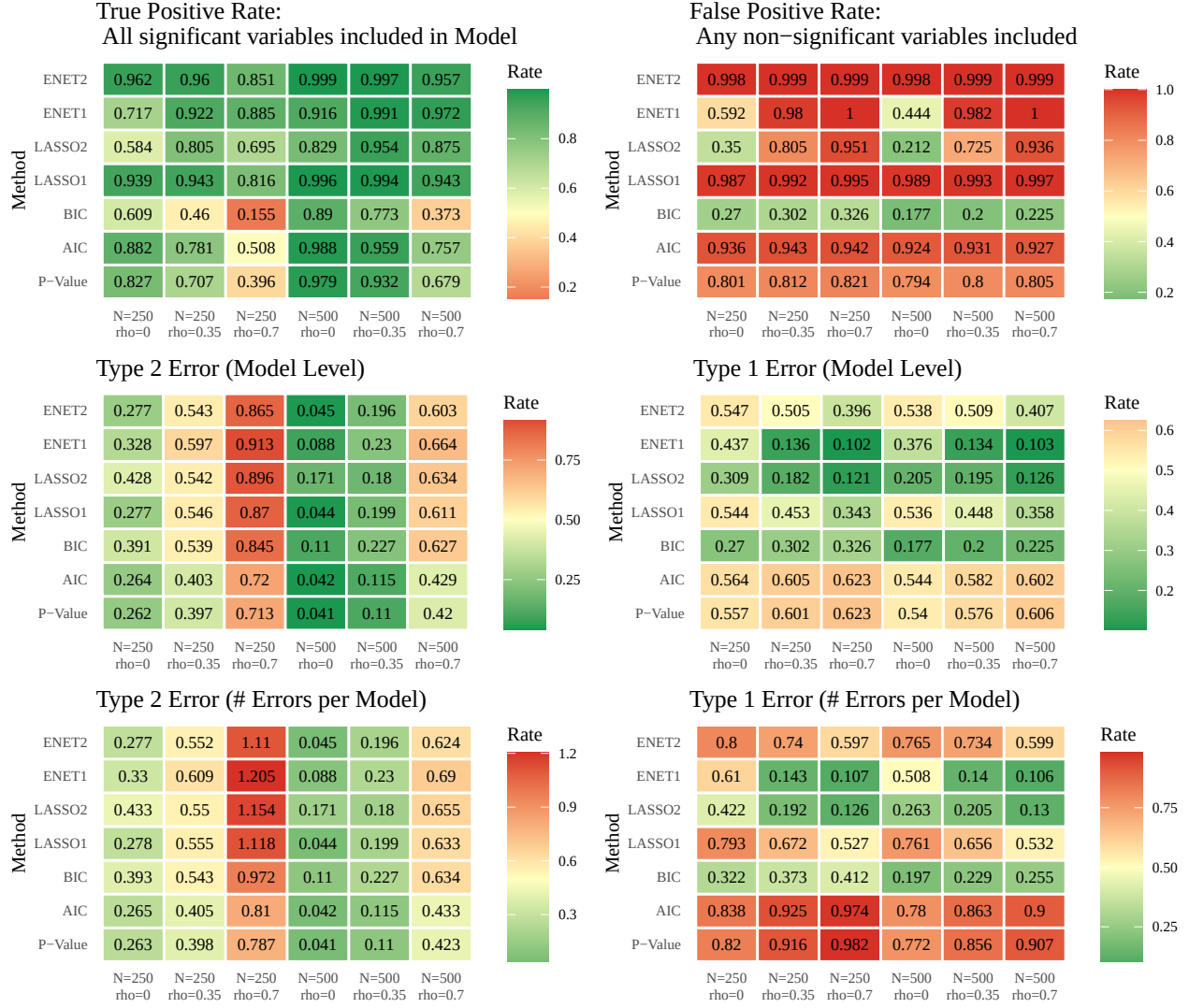


Figure 1: Heatmaps of simulation performance metrics across methods and scenarios.

Note: Top row displays true and false positive rates; middle row shows type I and II error rates at the model level; bottom row shows the average number of errors per model across methods and simulation scenarios. In every case, the gradient from red to green indicates worse to better performance

From Figure 1, true positive rates are highest for LASSO using λ_{min} and elastic net using λ_1 . However, these two methods also have the highest FPR. We see that AIC and p-value based methods have high false positive rates and type I error rates. Interestingly, BIC had

the lowest FPR consistently, and among the penalized regression methods, elastic net with λ_{min} and LASSO with λ_1 had lower FPR rates. However, as correlation increased, elastic net with λ_{min} and LASSO with λ_1 had better type 1 error rates than other approaches. Type 2 error was worse for penalized regression methods compared to backward selection methods in general, though TPR was better, especially for increasing correlation strength, meaning that the penalized regression models often kept all true variables in the model, though often without statistically significant estimates.

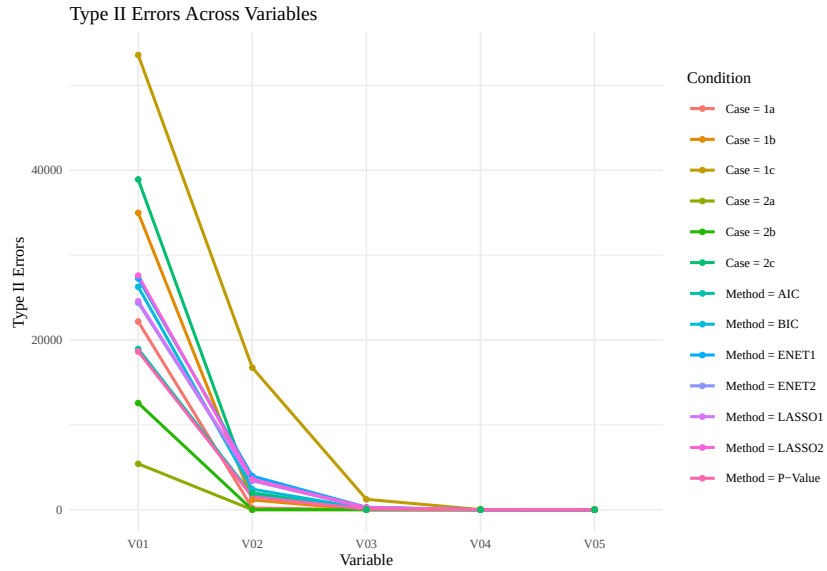


Figure 2: Type II error by magnitude of association, for methods and cases

The type II error trend as a variable's strength of association increases behaves as expected. Summed across both methods and cases, the weaker the association of a variable, the more type II errors were observed (Figure 2). Regarding bias and coverage of the null group, the penalized regression methods and BIC tended to have consistently stronger coverage, though the backward selection methods tended to have smaller bias (Table 3). Bias increased faster as correlation increased for penalized methods compared to backward selection methods,

Table 3: Average bias and coverage for null variables

Method	N = 250			N = 500		
	rho = 0	rho = 0.35	rho = 0.70	rho = 0	rho = 0.35	rho = 0.70
P-Value	0 (0.945)	-0.001 (0.939)	-0.002 (0.935)	0 (0.949)	0 (0.943)	-0.001 (0.94)
AIC	0 (0.944)	0 (0.938)	-0.002 (0.935)	0 (0.948)	0 (0.942)	-0.001 (0.94)
BIC	0 (0.979)	-0.001 (0.975)	-0.003 (0.973)	0 (0.987)	0 (0.985)	-0.001 (0.983)
LASSO1	0 (0.947)	-0.011 (0.955)	-0.017 (0.965)	0 (0.949)	-0.008 (0.956)	-0.012 (0.965)
LASSO2	0 (0.972)	-0.012 (0.987)	-0.02 (0.992)	0 (0.982)	-0.007 (0.986)	-0.014 (0.991)
ENET1	0 (0.959)	-0.016 (0.99)	-0.021 (0.993)	0 (0.966)	-0.011 (0.991)	-0.015 (0.993)
ENET2	0 (0.947)	-0.01 (0.951)	-0.016 (0.96)	0 (0.949)	-0.007 (0.951)	-0.011 (0.96)

Values reported as Bias (Coverage). Bias is the average coefficient estimate for null variables retained in the model (true value = 0). Coverage is calculated as 1 - (Type I error rate).

though penalized methods had better coverage. The shrinkage penalty in the penalization methods more often included unnecessary variables, but less often gave them an errantly significant effect. Relating to Figure 1, the shrinkage penalty often kept truly associated variables in the model, but more often failed to capture their true significance.

Conclusion

Across sample sizes and correlation strengths, a variety of trends were seen, and no method outperformed the others by every measure. Backward selection models are highly susceptible to type 1 error, which penalized regression methods handled better especially in the presence of multicollinearity. Penalized regression tended to have better coverage for weaker associations, especially with small N and large ρ but in general produced estimates more biased than those of backward selection methods. Among penalized regression models, LASSO with λ_{min} and elastic net with λ_1 performed comparatively to other methods in regard to TPR and type 2 error, but consistently mitigated type 1 error better and had the lowest bias across penalized regression models. Among backward selection models, BIC was more biased but handled

FPR and type 1 error much better than AIC or p-value methods. The most glaring trend that AIC and p-values have inflated type 1 error rates compared to other methods confirms the stigma against these methods in practice. Modelling is nuanced, and choosing an approach should be motivated by data structure and the hypotheses of interest, but penalized regression methods, especially LASSO with λ_{min} and elastic net with λ_1 , performed comparatively to, or even better than, other approaches across cases and performance measures, though BIC also handled bias and type 1 error well. Penalized regression may be a useful alternative to backwards selection especially in cases of multicollinearity and higher dimensional data, though variable selection should always be conducted with caution, and one should consider carefully the bias and error inflation it may induce.

Sources

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